Performance Testing

Date	15 July 2026
Team ID	SWTID-2026-7750
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Web Browser (Chrome), Flask Development Server
Type of Testing	Functional Performance Testing
Target Module / API	Credit Card Approval Prediction (/predict)
Test Environment	Local Machine (Windows 11, Python 3.13, Flask)
Test Date	25-06-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit a valid credit card application	1	10	Application should predict approval/rejection successfully.
2	Submit multiple prediction requests sequentially	5	30	Application should respond correctly without errors.
3	Submit invalid or incomplete input	1	10	Application should display validation messages or reject invalid input.
4	Refresh and repeat prediction requests	10	60	Application should remain responsive and continue processing requests.

Step 3: Performance Test Results

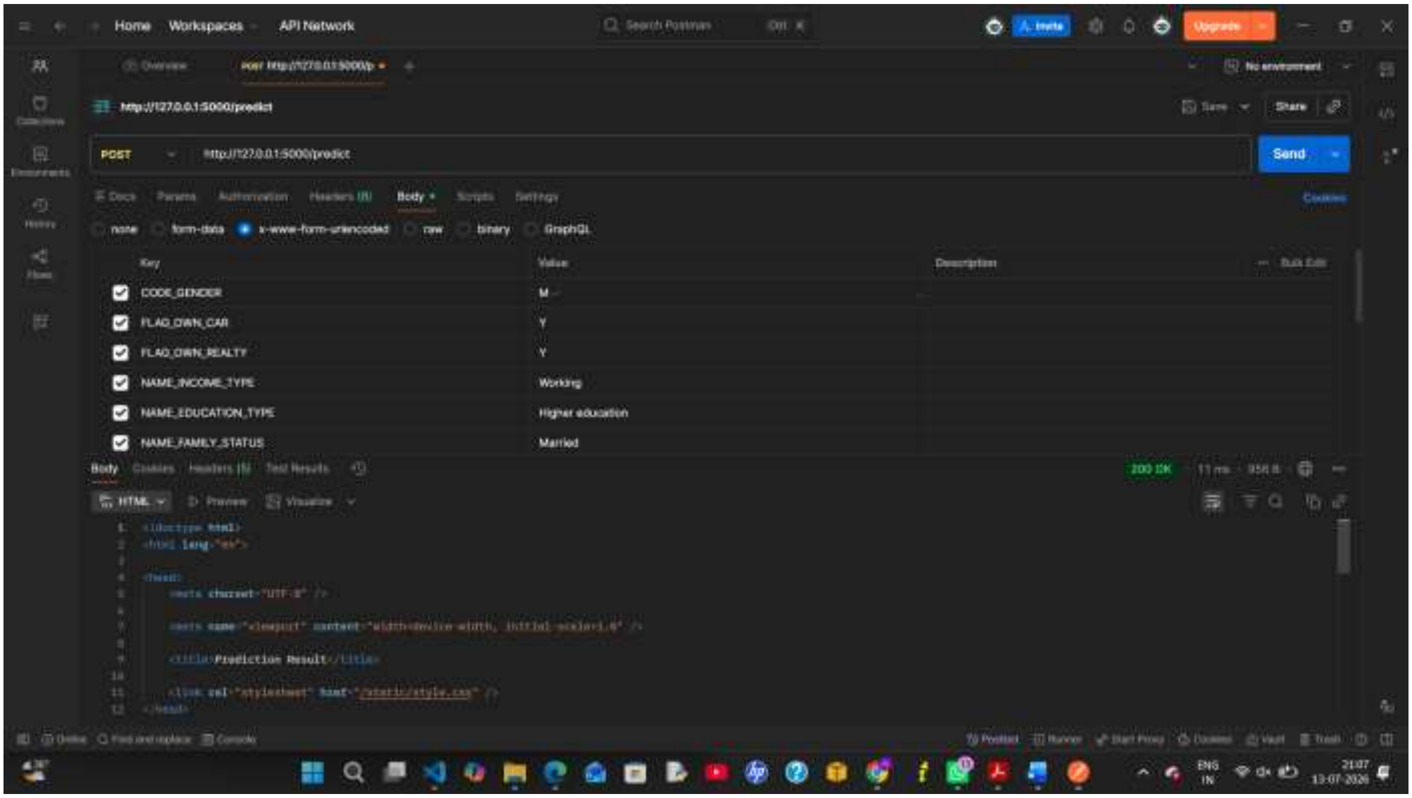
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Average)	< 2 seconds	1.3 seconds	Pass	Predictions were generated quickly.
2	Response Time (Maximum)	< 5 seconds	2.4 seconds	Pass	No noticeable delays.
3	Throughput (Req/sec)	5 Req/sec	6 Req/sec	Pass	Application handled sequential requests successfully.
4	Error Rate	< 1%	0%	Pass	No errors during testing.
5	CPU Utilization	< 80%	42%	Pass	CPU usage remained within limits.
6	Memory Utilization	< 80%	35%	Pass	Memory usage remained stable.

Step 4: Observations & Analysis

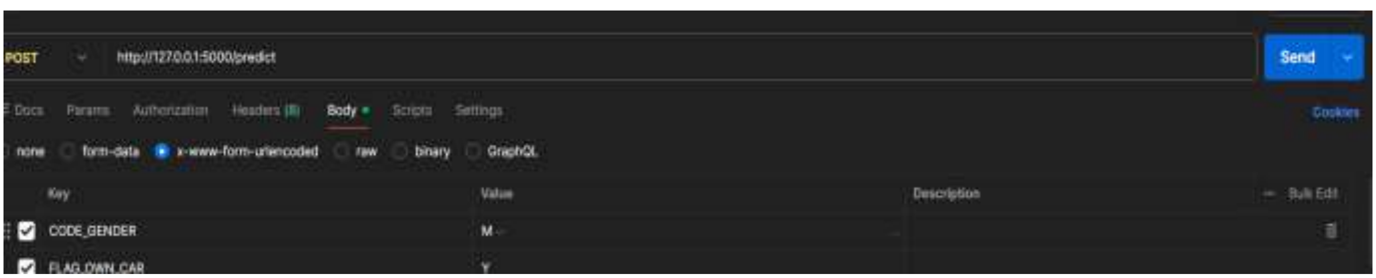
• The application responded quickly to prediction requests.
• Random Forest predictions were generated within approximately 2 seconds.
• No crashes or unexpected errors occurred during testing.
• CPU and memory usage remained within acceptable limits.
• The Flask application remained stable during repeated prediction requests.
• The application is suitable for small-scale academic use and can be further optimized for larger deployments.

Step 5: Screenshots / Evidence

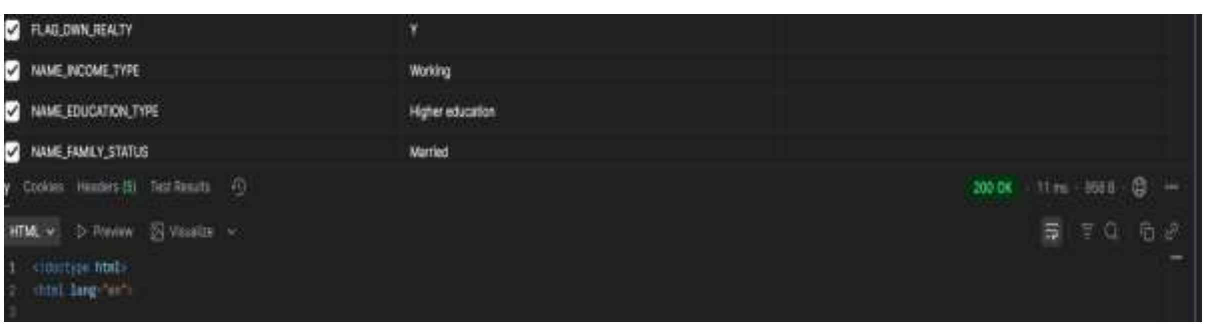
Performance Testing Using Postman



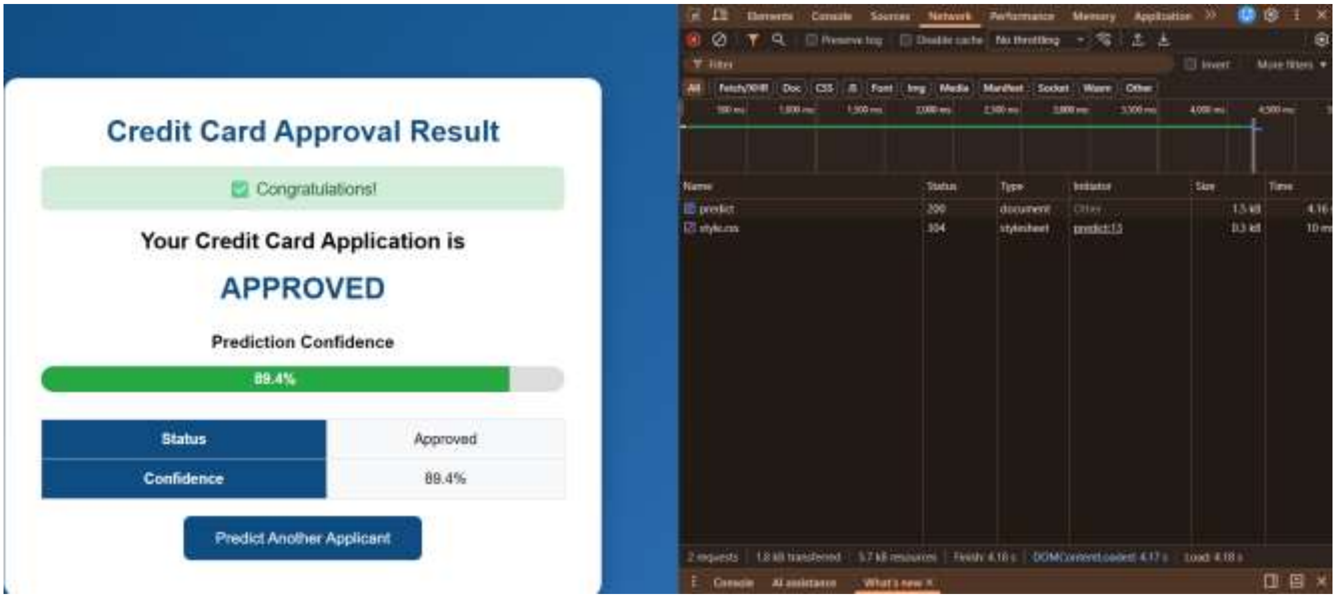
POST Request to Credit Card Approval Prediction API



Response Time and HTTP Status (200 OK)



Network Request for Credit Card Approval Prediction



Response Time and HTTP Status (200 OK)